It must be possible to shake an electric charge *here*, and to produce a wave which will drive an electric current *there*. Only a few years later, these waves, anticipated theoretically by Maxwell, will be revealed by the German physicist Heinrich Hertz; and just a few years later still, Guglielmo Marconi builds the first radio.

All modern communications technology – radio, television, telephones, computers, satellites, wi-fi, the internet, etc. – is an application of Maxwell's prediction; the Maxwell equations are the basis for all calculations made by telecommunications engineers. The contemporary world, based on communications, emerges from the intuitions of a poor London bookbinder – a skilful explorer of ideas with a vivid imagination – who saw some lines in his mind's eye; and the work of a good mathematician who translated this vision into equations, understanding that in the blink of an eye the waves of these lines can carry news from one side of the planet to the other.

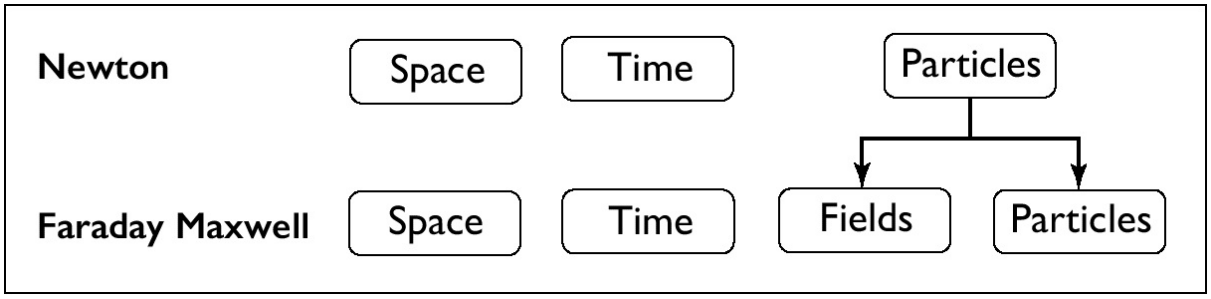


Figure 2.6 What is the world made of?

Our entire current technology is founded on the use of a physical thing – electromagnetic waves – which was not discovered empirically: it was predicted by Maxwell, simply by searching for the mathematical description accounting for the intuition Faraday got from bobbins and needles. This is the outstanding power of theoretical physics.

The world has changed: it is no longer made up of particles in space but of particles and fields in space (figure 2.6). It seems a minor change, but a few decades later a young Jewish man, a citizen of the world, will draw from it consequences that will go far beyond Michael Faraday's already fervid imagination, and which will shake Newton's world even further to its core.

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* $F = G\frac{M_1 M_2}{r^2}.$